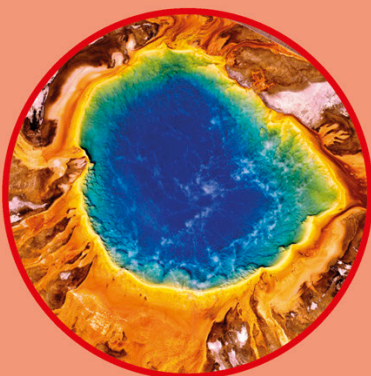




LIFE at *the*

NOT EVERYWHERE on our planet is a nice place to live. Some places are boiling hot, others are freezing cold, extremely salty, or have no oxygen. Yet there is always some life form that manages to survive there. We call these organisms *extremophiles*.

IN HOT WATER



HOT SPRING

In volcanic regions, where boiling water pours out of Earth's crust through geysers and vents, temperatures reach over 104°F (40°C) and the water is often full of sulphurous compounds. Although conditions are harsh, certain bacteria seem to thrive here. The green and red colors of the Grand Prismatic Spring in Yellowstone National Park are the result of bacteria growing around the edge of the pool.

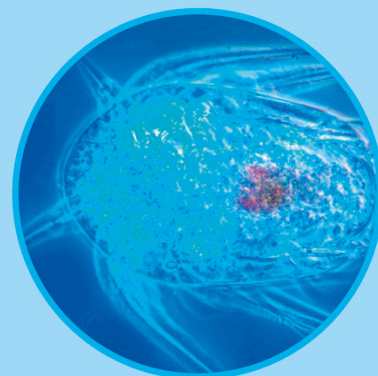
UNDER PRESSURE



RED TUBEWORM

On the deep seafloor, hydrothermal vents spew torrents of superheated, sulphur-rich water into the ocean. Temperatures inside the vents exceed 300°F (150°C) and the pressurized environment has no oxygen. Scientists have found microbes that can survive here, and not only that—the vents are also home to giant tubeworms, Pompeii worms, crabs, shrimp, and fish. They feed on the bacteria living in this poisonous environment.

FEELING CHILLY



MARINE PLANKTON

At the other end of the thermometer are organisms that live in icy conditions. These creatures have developed special proteins that act like antifreeze to prevent the fluid inside their cells turning to ice. Despite the cold, these areas of Earth are full of life. Polar waters contain bacteria that are food for larger krill, zooplankton, and fish. Some icefish are so sensitive that they would die from heatstroke if the temperature ever rose above 39°F (4°C).